CE334 - Laboratory 7
(2025-26 II Semester)

Generating BEM and nDSM using QGIS and LAStools

Objective

Generating BEM and nDSM and understanding these data products

Methodology

- Use the same file employed in the last exercise.
- Observe the point data in a viewer.
- Create a raster DSM of raw data where cell values are elevation values. Choose the size of the raster suitably.
    - Use the steps that are used to generate DSM in the previous lab.
- Generate a histogram of data i.e. frequency vs height of point. Observe the histogram.
    - Load the DSM layer (.tif file) into QGIS.
    - In the Layers panel located on leftside, you can see your loaded layer.
    - Right click on the layer and select properties.
    - Switch to histogram tab and compute the histogram.
- Decide a threshold on the basis of the raw data, raster image and histogram displays which can separate above ground objects from the ground.
    - Specify the threshold and write reason for selecting the threshold.
- Apply the threshold on the DSM image by converting above ground pixels as 9999 and the rest with their original values.
    a. Go to the Raster Tab in the menu toolbar and select the raster calculator.
    b. Type the expression:
        i. (raster_layer_name@1<threshold)*9999.
    c. Specify the output file path and its name.
    d. Hit OK.
    e. Try to do the steps (a) to (d) by loading the Raster file in MATLAB.
    (Specify what the white and black colour representing in this newly generated dsm along with the reason.)
- Remove each 9999 pixel by interpolation from neighbouring pixels (i.e. ground pixels).
    - Load the .tif file generated after changing the pixel values into MATLAB using imread function.
    - You can see that your data is stored in the form of matrix. Write the dimensions of the matrix.
    - Open the variable in which the image is stored and take a look on its values. You can see that some cells have a value 9999 which you have changed earlier.
    - You can also see some cells which have a large negative value.
    - Display the loaded image using “imagesc(variable)”.

- Then try to change the values which are less than 0 in the matrix to zero.
- Then again, view the image and write the changes you observed in the image and attach the loaded image before and after changing the values in the report.
- Then create a mask of the pixel values which are equal to 9999 (remember cells in the matrix represent the pixels of your image.)
- Then, Use the region fill command to remove the pixels with 9999 values and fill it with interpolated values.
- *This is just one way to achieve this. Try to find other ways of interpolating the neighbouring pixel values into the pixels which are non-ground pixels.*

• This is the BEM. Open the variable in which you stored the interpolated values and try to compare the height values of the non-ground pixels.

• Load the DSM file in the matlab.

• Find nDSM = DSM – BEM (Since both the variables are matrices, so basic matrix operations apply here).

• Display nDSM in matlab image viewer and check the pixel values.

• Write the change in the ground pixel values after subtraction and why is this image called as nDSM.

Report

Your report should contain the following:

• Images of original data.

• DSM and nDSM images created using the same vertical scale. Try to understand why we want the same vertical scale for these two data sets and write in report.

• Your observation on this method. Also, your comment on how BEM and nDSM can be generated by directly working with point cloud and not converting these to raster.

• Also, write in what kind of terrain this method will fail.